

Olasılıksal Robotik

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Eşanlı Konum Belirleme ve Haritalama - SLAM

- Full SLAM

$$p \left(x_{1:t}, m \mid z_{1:t}, u_{1:t} \right)$$

- Online SLAM

$$p \left(x_t, m \mid z_{1:t}, u_{1:t} \right)$$

Eşanlı Konum Belirleme ve Haritalama - SLAM

- Full SLAM with correspondence variable

$$p \left(x_{1:t}, m, c_t \mid z_{1:t}, u_{1:t} \right)$$

- Online SLAM with correspondence variable

$$p \left(x_t, m, c_t \mid z_{1:t}, u_{1:t} \right)$$

EKF SLAM

- EKF SLAM =
 - EKF tabanlı hareket modeli
 - EKF tabanlı sensör modeli
 - Online SLAM
 - Ençok Olabilirlik veri ilişkilendirme(Maximum Likelihood data association)

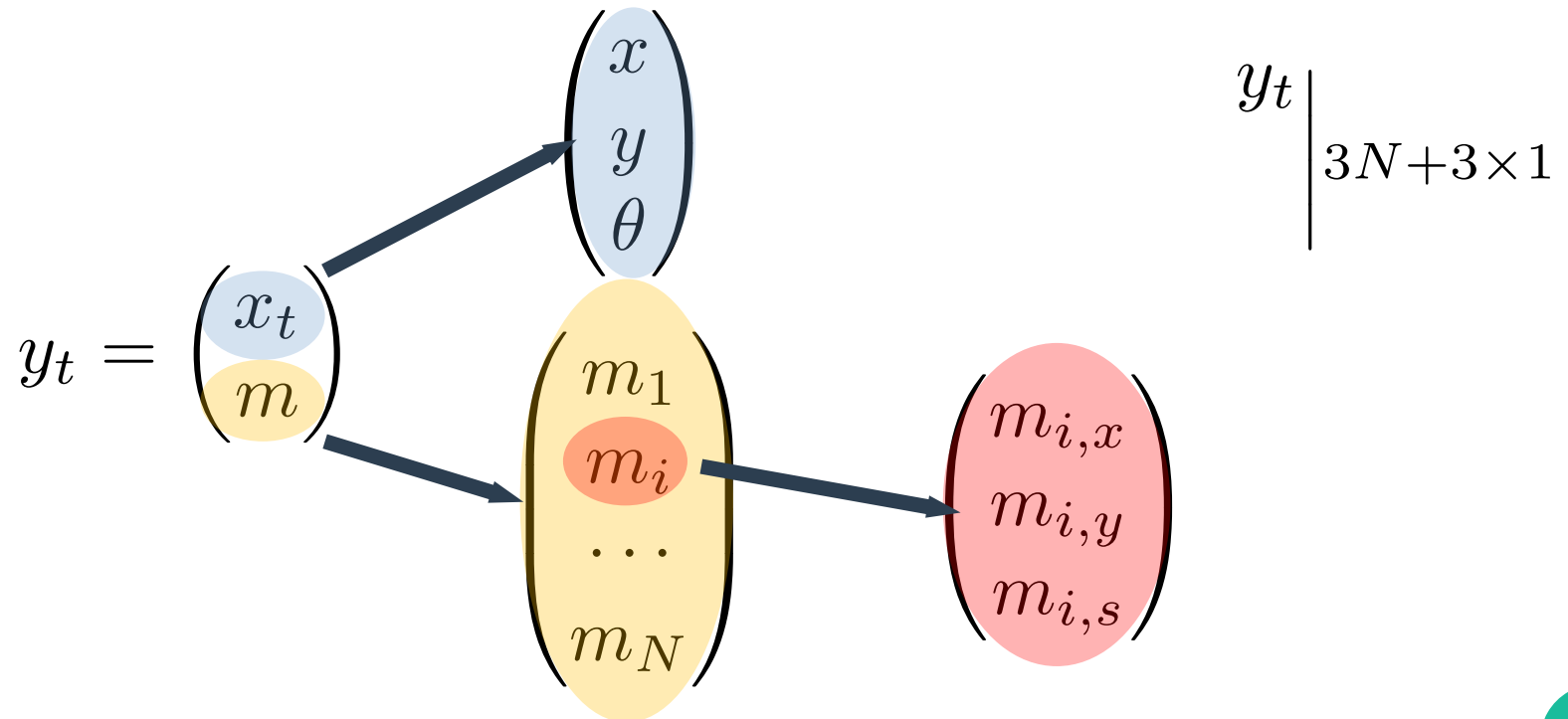
EKF SLAM

- EKF SLAM =
 - Özellik tabanlı haritalar (Feature based map)
 - Noktasal mihenktaşları (point landmarks)
 - Toplam landmark sayısı < 1000
 - Hareket ve sensör modeli için Gauss gürültüsü varsayımı

EKF SLAM – Bilinen ölçüm mihenktaşı ilişkisi

EKF – Known Correspondence

- EKF konum belirleme ile büyük benzerlik taşır
- Durum uzayına (y_t), robot konumuna (x_t) ek olarak landmarkların konum bilgisi (m) de eklenir



EKF SLAM - ilklendirme

$$\mu_0 = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}_{(3N+3) \times 1}$$

$$\Sigma_0 = \begin{pmatrix} 0 & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \infty & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \dots & \infty \end{pmatrix}_{(3N+3) \times (3N+3)}$$

EKF SLAM – Hareket Modeli

$$y_t = y_{t-1} + \begin{pmatrix} -\frac{v_t}{\omega_t} \sin(\theta) + \frac{v_t}{\omega_t} \sin(\theta + \omega_t \Delta_t) \\ \frac{v_t}{\omega_t} \cos(\theta) - \frac{v_t}{\omega_t} \cos(\theta + \omega_t \Delta_t) \\ \omega_t \Delta_t + \gamma \Delta_t \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

EKF SLAM – Hareket Modeli

$$F_x = \begin{pmatrix} 1 & 0 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & 1 & 0 & \dots & 0 \end{pmatrix}_{(3) \times (3N+3)}$$

$$y_t = y_{t-1} + F_x^T \begin{pmatrix} -\frac{v_t}{\omega_t} \sin(\theta) + \frac{v_t}{\omega_t} \sin(\theta + \omega_t \Delta_t) \\ \frac{v_t}{\omega_t} \cos(\theta) - \frac{v_t}{\omega_t} \cos(\theta + \omega_t \Delta_t) \\ \omega_t \Delta_t + \gamma \Delta_t \end{pmatrix}$$

$$y_t = y_{t-1} + \underbrace{F_x^T \begin{pmatrix} -\frac{v_t}{\omega_t} \sin(\theta) + \frac{v_t}{\omega_t} \sin(\theta + \omega_t \Delta_t) \\ \frac{v_t}{\omega_t} \cos(\theta) - \frac{v_t}{\omega_t} \cos(\theta + \omega_t \Delta_t) \\ \omega_t \Delta_t + \gamma \Delta_t \end{pmatrix}}_{g(u_t, y_{t-1})} + \mathcal{N}(0, F_x^T R_t F_x)$$

EKF SLAM – Hareket Modeli

$$g(u_t, y_{t-1}) \approx g(\mu_{u_t}, \mu_{t-1}) + G_t(y_{t-1}, \mu_{t-1})$$

$$G_t = \left. \frac{\partial g(u_t, y_{t-1})}{\partial y_{t-1}} \right|_{u_t, y_{t-1} = \mu_{t-1}}$$

$$G_t = g'(u_t, \mu_{t-1})$$

$$G_t = I + F_x^T g_t F_x$$

$$g_t = \begin{pmatrix} 0 & 0 & -\frac{v_t}{\omega_t} \cos(\mu_{t-1, \theta}) + \frac{v_t}{\omega_t} \cos(\mu_{t-1, \theta} + \omega_t \Delta_t) \\ 0 & 0 & -\frac{v_t}{\omega_t} \sin(\mu_{t-1, \theta}) + \frac{v_t}{\omega_t} \sin(\mu_{t-1, \theta} + \omega_t \Delta_t) \\ 0 & 0 & 0 \end{pmatrix}$$

EKF SLAM – Hareket Modeli

$$\bar{\mu}_t = \mu_{t-1} + F_x^T \begin{pmatrix} -\frac{v_t}{\omega_t} \sin(\mu_{t-1, \theta}) + \frac{v_t}{\omega_t} \sin(\mu_{t-1, \theta} + \omega_t \Delta_t) \\ \frac{v_t}{\omega_t} \cos(\mu_{t-1, \theta}) - \frac{v_t}{\omega_t} \cos(\mu_{t-1, \theta} + \omega_t \Delta_t) \\ \omega_t \Delta_t \end{pmatrix}$$

$$\bar{\Sigma}_t = G_t \Sigma_{t-1} G_t^T + F_x^T R_t F_x$$

EKF SLAM – Hareket Modeli

Algorithm EKF_SLAM_known_correspondences($\mu_{t-1}, \Sigma_{t-1}, u_t, z_t, c_t$):

$$F_x = \begin{pmatrix} 1 & 0 & 0 & 0 \dots 0 \\ 0 & 1 & 0 & 0 \dots 0 \\ 0 & 0 & 1 & \underbrace{0 \dots 0}_{3N} \end{pmatrix}$$

$$\bar{\mu}_t = \mu_{t-1} + F_x^T \begin{pmatrix} -\frac{v_t}{\omega_t} \sin \mu_{t-1,\theta} + \frac{v_t}{\omega_t} \sin(\mu_{t-1,\theta} + \omega_t \Delta t) \\ \frac{v_t}{\omega_t} \cos \mu_{t-1,\theta} - \frac{v_t}{\omega_t} \cos(\mu_{t-1,\theta} + \omega_t \Delta t) \\ \omega_t \Delta t \end{pmatrix}$$

$$G_t = I + F_x^T \begin{pmatrix} 0 & 0 & -\frac{v_t}{\omega_t} \cos \mu_{t-1,\theta} + \frac{v_t}{\omega_t} \cos(\mu_{t-1,\theta} + \omega_t \Delta t) \\ 0 & 0 & -\frac{v_t}{\omega_t} \sin \mu_{t-1,\theta} + \frac{v_t}{\omega_t} \sin(\mu_{t-1,\theta} + \omega_t \Delta t) \\ 0 & 0 & 0 \end{pmatrix} F_x$$

$$\bar{\Sigma}_t = G_t \Sigma_{t-1} G_t^T + F_x^T R_t F_x$$

EKF SLAM – Sensör Modeli

$$Q_t = \begin{pmatrix} \sigma_r^2 & 0 & 0 \\ 0 & \sigma_\phi^2 & 0 \\ 0 & 0 & \sigma_s^2 \end{pmatrix}$$

for all observed features $z_t^i = (r_t^i \ \phi_t^i \ s_t^i)^T$ do

$$j = c_t^i$$

if landmark j never seen before

$$\begin{pmatrix} \bar{\mu}_{j,x} \\ \bar{\mu}_{j,y} \\ \bar{\mu}_{j,s} \end{pmatrix} = \begin{pmatrix} \bar{\mu}_{t,x} \\ \bar{\mu}_{t,y} \\ s_t^i \end{pmatrix} + \begin{pmatrix} r_t^i \cos(\phi_t^i + \bar{\mu}_{t,\theta}) \\ r_t^i \sin(\phi_t^i + \bar{\mu}_{t,\theta}) \\ 0 \end{pmatrix}$$

endif

$$\delta = \begin{pmatrix} \delta_x \\ \delta_y \end{pmatrix} = \begin{pmatrix} \bar{\mu}_{j,x} - \bar{\mu}_{t,x} \\ \bar{\mu}_{j,y} - \bar{\mu}_{t,y} \end{pmatrix}$$

$$q = \delta^T \delta$$

$$\hat{z}_t^i = \begin{pmatrix} \sqrt{q} \\ \text{atan2}(\delta_y, \delta_x) - \bar{\mu}_{t,\theta} \\ \bar{\mu}_{j,s} \end{pmatrix}$$

EKF SLAM – Sensör Modeli

$$F_{x,j} = \begin{pmatrix} 1 & 0 & 0 & \underbrace{0 \dots 0}_{3j-3} & 0 & 0 & 0 & \underbrace{0 \dots 0}_{3N-3j} \\ 0 & 1 & 0 & 0 \dots 0 & 0 & 0 & 0 & 0 \dots 0 \\ 0 & 0 & 1 & 0 \dots 0 & 0 & 0 & 0 & 0 \dots 0 \\ 0 & 0 & 0 & 0 \dots 0 & 1 & 0 & 0 & 0 \dots 0 \\ 0 & 0 & 0 & 0 \dots 0 & 0 & 1 & 0 & 0 \dots 0 \\ 0 & 0 & 0 & 0 \dots 0 & 0 & 0 & 1 & 0 \dots 0 \end{pmatrix}$$

$$H_t^i = \frac{1}{q} \begin{pmatrix} -\sqrt{q}\delta_x & -\sqrt{q}\delta_y & 0 & +\sqrt{q}\delta_x & \sqrt{q}\delta_y & 0 \\ \delta_y & -\delta_x & -q & -\delta_y & +\delta_x & 0 \\ 0 & 0 & 0 & 0 & 0 & q \end{pmatrix} F_{x,j}$$

$$K_t^i = \bar{\Sigma}_t H_t^{iT} (H_t^i \bar{\Sigma}_t H_t^{iT} + Q_t)^{-1}$$

$$\bar{\mu}_t = \bar{\mu}_t + K_t^i (z_t^i - \hat{z}_t^i)$$

$$\bar{\Sigma}_t = (I - K_t^i H_t^i) \bar{\Sigma}_t$$

endfor

$$\mu_t = \bar{\mu}_t$$

$$\Sigma_t = \bar{\Sigma}_t$$

return μ_t, Σ_t